

ASG MASTER AUTOCAD TEMPLATE - CAD STANDARDS & SETTINGS REFERENCE

AL SIRAJ GROUP | Aluminium & Glass Division | UAE | Rev 0 | 10.10.2026

Files: ASG_Master_Template.dxf (master, AutoCAD 2018 DXF), ASG_Monochrome.ctb, ASG_Template_Setup.lsp, ASG_Layer_Standards.xlsx/.csv, ASG_Validation_Report.txt.
ASG_Master_Template.dwg/.dwt must be saved from AutoCAD (see section 14) - no DWG writer exists in the generation environment.

1. Drawing units & drawing-resident system variables

Variable	Value	Purpose
INSUNITS	4 (Millimetres)	Block / xref insertion units
MEASUREMENT	1 (Metric)	ISO hatch & linetype files
LUNITS / LUPREC	2 Decimal / 0	Ordinary lengths whole mm (raise per detail via dim style)
AUNITS / AUPREC	0 Decimal degrees / 2	Angles 0.00 deg
ANGBASE / ANGDIR	0 / 0	East = 0, counter-clockwise
LIMMIN / LIMMAX	0,0 / 100000,70000	Nominal area; LIMCHECK = 0 (no restriction)
LTSCALE / PSLTSCALE	1 / 1	Linetypes in paper mm
LWDISPLAY	1	Lineweights shown on screen
CELWEIGHT / CECOLOR	ByLayer / ByLayer	No hard-coded properties
CLAYER	ASG-OBJECT	Default current layer
TEXTSTYLE / DIMSTYLE	ASG-TEXT / ASG-DIM-ANNOTATIVE	Defaults
CMLSTYLE	ASG-MLINE-2	Default multiline style
ORTHOMODE	0	Use polar tracking instead
DIMASSOC	2	Associative dimensions - follow real geometry
PLINEGEN / FILLMODE / MIRRTEXT	1 / 1 / 0	Linetype along polylines; fills on; text not mirrored
PSTYLEMODE	1	Colour-dependent plot styles (CTB)
Model space	Empty, World UCS	No geometry, no xrefs

2. Machine / profile settings (NOT stored in DWG/DWT)

These live in the AutoCAD user profile / registry. They are NOT written by the template. Recommended values below; ASG_Template_Setup.lsp contains an optional (ASG-PROFILE) command that sets them only when a user runs it.

Setting	Recommended	Note
OSMODE	2223	End, Mid, Centre, Node, Intersection, Perpendicular, Extension (registry, not DWG)
AUTOSNAP	63	Marker, magnet, tooltip, aperture, OTRACK
POLARMODE / POLARANG	Tracking ON / 45	Polar + object snap tracking
DYNMODE / DYNPROMPT	3 / 1	Dynamic input on
SELECTIONCYCLING	2	Cycle overlapping objects

Setting	Recommended	Note
GRIPS / GRIPSIZE	1 / 5	Standard grips
PICKBOX / APERTURE	4 / 8	
PICKFIRST / PICKADD	1 / 2	Noun-verb, Shift to remove
MSLTSCALE / ANNOAUTOSCALE	1 / 0	Annotation linetype scaling; no auto scale add
SAVETIME / ISAVEBAK	10 min / 1	Autosave + .bak
Background / visual style	Dark grey 33,40,48 model; white paper / 2D Wireframe	Options > Display
REGENAUTO / LAYOUTREGENCTL	On / 2	Fast layout switching
HPLAYER / DIMLAYER	ASG-HATCH / ASG-DIMENSION	Auto layers (AutoCAD 2016+)

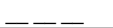
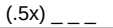
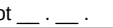
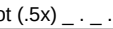
3. Layer standard

Layer	ACI	Linetype	LW	Plot	Purpose
ASG-OBJECT	7	Continuous	0.25	Yes	General object geometry - default current layer
ASG-OUTLINE	7	Continuous	0.35	Yes	Main overall outlines / visible edges in elevation
ASG-PROFILE	7	Continuous	0.30	Yes	Aluminium profile outlines (frames, mullions, transoms, sashes)
ASG-GLASS	4	Continuous	0.18	Yes	Glass / mirror panel edges
ASG-METAL	7	Continuous	0.30	Yes	MS / SS members, plates, brackets
ASG-HARDWARE	8	Continuous	0.18	Yes	Handles, hinges, locks, rollers, patch fittings
ASG-FIXING	8	Continuous	0.18	Yes	Anchors, screws, bolts, fixing brackets
ASG-CLADDING	7	Continuous	0.25	Yes	ACP / panel cladding and flashings
ASG-INSULATION	252	Continuous	0.13	Yes	Insulation / thermal break outline and hatch (plots grey)
ASG-HIDDEN	8	HIDDEN2	0.18	Yes	Concealed / hidden edges
ASG-CENTER	8	CENTER2	0.13	Yes	Centrelines and axes of symmetry
ASG-SECTION	7	Continuous	0.35	Yes	Cut-section outlines (heaviest object line)
ASG-HATCH	252	Continuous	0.09	Yes	General section hatching (plots grey)
ASG-GLAZING-BEAD	7	Continuous	0.25	Yes	Glazing beads and snap-in profiles
ASG-SEALANT	8	Continuous	0.13	Yes	Silicone / weather seal, backer rod
ASG-GASKET	8	Continuous	0.13	Yes	EPDM gaskets, brushes, setting blocks
ASG-DRAINAGE	8	DASHED2	0.18	Yes	Drainage paths, weep holes, linear drains
ASG-CONSTRUCTION	6	Continuous	0.05	NO	Construction / setting-out lines - NO PLOT
ASG-REFERENCE	253	Continuous	0.13	Yes	Architectural / structural background (plots light grey)
ASG-DIMENSION	3	Continuous	0.13	Yes	Dimensions
ASG-TEXT	7	Continuous	0.18	Yes	General text
ASG-TEXT-NOTE	7	Continuous	0.18	Yes	General / specification notes
ASG-MULTILEADER	3	Continuous	0.13	Yes	Multileader callouts
ASG-TABLE	7	Continuous	0.18	Yes	Schedules and tables
ASG-LEVEL	3	Continuous	0.18	Yes	Level markers (FFL, SSL, sill heights)
ASG-REVISION	1	Continuous	0.25	Yes	Revision clouds and delta tags
ASG-DETAIL-MARK	3	Continuous	0.25	Yes	Detail callout bubbles / boundaries

Layer	ACI	Linetype	LW	Plot	Purpose
ASG-SECTION-MARK	3	Continuous	0.25	Yes	Section cut lines and markers
ASG-GRID	8	CENTER	0.13	Yes	Column grid / setting-out grid
ASG-TITLEBLOCK	7	Continuous	0.25	Yes	Title block linework and attributes
ASG-BORDER	7	Continuous	0.50	Yes	Sheet border frame
ASG-VIEWPORT	6	Continuous	0.00	NO	Viewport boundaries - NO PLOT
ASG-NO PLOT	6	Continuous	0.00	NO	Reference notes / markups never printed - NO PLOT

Colour logic: 7 primary geometry and text | 8 secondary, hidden, hardware | 4 glass | 3 annotation (dims, leaders, marks) | 1 revision | 6 non-plot | 250-254 deliberately plotted grey (hatch, insulation, reference). Default current layer: ASG-OBJECT. Layer 0 = blocks only. Defpoints is created automatically by AutoCAD with the first dimension - not pre-created.

4. Linetypes & lineweight hierarchy

Linetype	Description
HIDDEN	Hidden 
HIDDEN2	Hidden (.5x) 
CENTER	Center 
CENTER2	Center (.5x) 
DASHED	Dashed 
DASHED2	Dashed (.5x) 
DASHDOT	Dash dot 
DASHDOT2	Dash dot (.5x) 
PHANTOM	Phantom 
PHANTOM2	Phantom (.5x) 
DOT	Dot 
Continuous	Solid
Lineweight	Class / use
0.50 mm	Sheet border
0.35 mm	Main outline / cut section
0.30 mm	Profiles / metal
0.25 mm	Secondary geometry / marks
0.18 mm	Glass, hidden, hardware, text
0.13 mm	Dimensions, centrelines, seals
0.09 mm	Hatch
0.05 mm	Construction (no plot)

LTSCALE = 1, PSLTSCALE = 1: linetype dash lengths are paper mm in every viewport. MSLTSCALE (profile, set 1) scales them by the annotation scale in model space. Use CELTSCALE only per object if a short member needs a tighter pattern.

5. Text styles - font Arial (TrueType, present on every Windows PC)

Style	Font	Annotative	Height in style	Use
ASG-TEXT	arial.ttf	Yes	0 (variable)	General text and dimension text
ASG-TEXT-NOTE	arial.ttf	Yes	0 (variable)	General notes / specifications
ASG-TEXT-SMALL	arial.ttf	Yes	0 (variable)	Small technical annotation, profile codes
ASG-TEXT-TITLE	arialbd.ttf	No (paper space)	0 (variable)	Detail / view titles (5.0 for drawing title)
ASG-TEXT-SHEET	arialbd.ttf	No (paper space)	0 (variable)	Sheet headings, title block title / number

Text use	Paper height	Style	Note
Small technical annotation	1.8 mm	ASG-TEXT-SMALL	Min. 1.8 mm - not below on A4
Dimensions	2.5 mm	ASG-TEXT (via dim style)	2.0 acceptable on congested 1:1 details
General notes	2.5 mm	ASG-TEXT-NOTE	
Detail / view titles	3.5 mm	ASG-TEXT-TITLE	Underlined, scale text 2.5 below
Drawing titles	5.0 mm	ASG-TEXT-SHEET	7.0 on A1 title block
Sheet headings	5.0 mm	ASG-TEXT-SHEET	3.5 on A4

Readable on A4, A3 and A1: minimum 1.8 mm; never reduce title block text when plotting A1 to A3 half-size - plot A3 at full size from an A3 layout instead.

6. Dimension styles

Style	Annotative	DIMSCALE	Use
ASG-DIM-ANNOTATIVE	Yes	1 (scale from annotation scale)	Default - model space, any viewport scale
ASG-DIM-1-1	No	1	Fixed scale 1:1 model-space dimensions
ASG-DIM-1-2	No	2	Fixed scale 1:2 model-space dimensions
ASG-DIM-1-5	No	5	Fixed scale 1:5 model-space dimensions
ASG-DIM-1-10	No	10	Fixed scale 1:10 model-space dimensions
ASG-DIM-1-20	No	20	Fixed scale 1:20 model-space dimensions
ASG-DIM-1-50	No	50	Fixed scale 1:50 model-space dimensions

Parameter	Value
Text style / height	ASG-TEXT, 2.5 mm, above line, aligned with line, gap 1.0
Arrowhead	Closed filled 2.5 mm; centre mark 2.5
Extension lines	Offset from origin 1.5, extend beyond dim line 1.25
Baseline spacing (DIMDLI)	7.0
Fit	Either text or arrows (best fit), text beside line when moved, fit always inside ext lines
Units / precision	Decimal, 0 decimals, '.' separator, trailing zeros suppressed; angles 0.00 deg
Special details	Override precision in a child style or per-detail style (e.g. 0.0 for 1:1 fabrication)
Measurement scale (DIMLFAC)	1.0 - always true measured geometry, never type over values
Tolerance	None by default; symmetrical +/- precision 0.0, height 0.7 when needed
Alternate units	Off (inches 25.4 preset, 2 dp)
Colour / lineweight	ByBlock inside style -> ByLayer from ASG-DIMENSION

7. Multileader styles

Style	Annotative	Arrow	Text	Landing / dogleg	Use
ASG-ML-ANNOTATIVE	Yes	Closed filled 2.5	ASG-TEXT-NOTE 2.5	gap 1.0 / 4.0	Default callouts in model space
ASG-ML-NOTE	No	Closed filled 2.5	ASG-TEXT-NOTE 2.5	gap 1.0 / 4.0	Paper-space notes
ASG-ML-DETAIL	No	Dot 1.5	ASG-TEXT-SMALL 1.8	gap 1.0 / 4.0	Dense 1:1 / 1:2 details, paper space

All: straight leader, max 2 points, MText content, no text frame, attachment middle of top line both sides, colour/lineweight ByLayer. Background mask is an MText property, not a style property - turn on per note (Properties > Text > Background mask) where a note sits over hatch.

8. Annotative strategy & annotation scales

Scale list (ACAD_SCALELIST): 1:1, 1:2, 1:5, 1:10, 1:20, 1:25, 1:50, 1:100 - no duplicates, no imperial scales.

Object	Approach
Text (model space)	ASG-TEXT / -NOTE / -SMALL are annotative - type paper height
Text (paper space)	Titles / sheet text: ASG-TEXT-TITLE / -SHEET, non-annotative, real paper height
Dimensions	ASG-DIM-ANNOTATIVE in model space; fixed ASG-DIM-1-x only for single-scale drawings
Multileaders	ASG-ML-ANNOTATIVE in model space; ASG-ML-NOTE / -DETAIL in paper space
Blocks	Annotative only for symbols (section/detail/level marks); real-size items never
Hatches	Non-annotative for material sections (true size); annotative only for symbolic fills

Manage scales: set the viewport scale first (select viewport > scale list on status bar). Annotation added in model space while CANNOSCALE = viewport scale gets that scale. To show an object in a second viewport scale: select it > right-click > Annotative Object Scale > Add/Delete (OBJECTSCALE). Keep ANNOAUTOSCALE = 0 (do not auto-add every scale). SCALELISTEDIT > Reset only by the CAD custodian; run -SCALELISTEDIT > Reset > Exit then AUDIT if imported drawings bring _XREF or duplicate scales.

9. Hatch standard (standard acadiso.pat patterns only)

Material	Pattern	Scale @1:1	Angle	Layer	Note
Glass (section)	ANSI31	0.5	0	ASG-HATCH	Or no hatch + 2 parallel lines for thin glass
Aluminium section	SOLID	-	-	ASG-HATCH	Thin walls: solid fill reads best
Steel / metal section	ANSI32	1.0	0	ASG-HATCH	SS: ANSI31 @ 90 to distinguish
Concrete	AR-CONC	0.05	0	ASG-REFERENCE	Scale x drawing scale
Masonry / blockwork	ANSI31	1.0	0	ASG-REFERENCE	Alt: AR-B816 in elevation
Insulation	INSUL	0.05	0	ASG-INSULATION	Fit to insulation thickness
Sealant	SOLID	-	-	ASG-SEALANT	Small beads: solid fill
Generic section fill	ANSI37	1.0	0	ASG-HATCH	

Model-space hatch scale = table scale x drawing scale (e.g. ANSI31 at 1:5 = 1.0 x 5), or tick Annotative in the hatch dialog and leave scale at the table value. HPLAYER = ASG-HATCH (profile/session). The master drawing contains no hatches.

10. Multiline styles

ASG-MLINE-2 (two lines +/-0.5, square caps) and ASG-MLINE-3 (adds CENTER2 axis). MLINE justification Zero, scale = overall width (e.g. 50 for a 50 mm frame).

11. Table styles (created by ASG_Template_Setup.lsp - UNVERIFIED)

Style	Title row	Header row	Data rows	Use
ASG-TABLE-STANDARD	3.5 bold, centred	2.5 bold, centred, grey fill	2.5, middle-left	General / drawing register
ASG-TABLE-MATERIAL	3.5 bold, centred	2.0 bold, centred, grey fill	2.0, middle-left	Material, glass, hardware schedules
ASG-TABLE-REVISION	suppressed	2.0 bold, centred	2.0, middle-left	Revision history (Rev, Date, Description, By)
ASG-TABLE-NOTES	3.5 bold, left	suppressed	2.5, top-left	General notes blocks

Common: text style ASG-TEXT, horizontal margin 1.5, vertical margin 1.0, direction down, grid 0.18 inside / 0.35 outline, colours ByBlock, no background fill on data rows, light grey (ACI 254) fill on header row.

12. Block & attribute standard

Naming: ASG-[CATEGORY]-[ITEM]-[VARIANT] (e.g. ASG-HW-HINGE-01). Draw on layer 0 with ByBlock/ByLayer properties, base point at a logical fixing/datum point (bottom-left or centreline), unit 1 = 1 mm, Insert units = Millimetres, 'Scale uniformly' ON, 'Allow exploding' OFF for symbols. Symbols that must stay a fixed paper size (section/detail marks, level heads, north arrow) are Annotative; real-size product blocks are not. Attribute definitions on layer 0, style ASG-TEXT-SMALL, invisible unless displayed.

Tag	Prompt	Example
ITEM_CODE	Item / product code	e.g. ASG-W-01
ITEM_NAME	Item description	e.g. SLIDING WINDOW
SYSTEM_NAME	System / series	e.g. supplier series name
MANUFACTURER	Manufacturer / supplier	
MATERIAL	Base material	ALUMINIUM 6063-T6 / SS304 ...
FINISH	Finish	POWDER COATED / ANODISED / HAIRLINE
COLOUR	Colour / RAL	RAL 9016
GLASS_SPEC	Glass specification	6mm CLEAR TEMPERED
REVISION	Block revision	R0

13. Layouts, page setups, title block & plotting

Layout	Paper (DWG To PDF.pc3)	Title block	Viewport
A4-LANDSCAPE	ISO_full_bleed_A4_(297.00_x_210.00_MM)	ASG-TB-A4	1 locked, 1:10, layer ASG-VIEWPORT
A4-PORTRAIT	ISO_full_bleed_A4_(210.00_x_297.00_MM)	ASG-TB-A4	1 locked, 1:10, layer ASG-VIEWPORT
A3-LANDSCAPE	ISO_full_bleed_A3_(420.00_x_297.00_MM)	ASG-TB-A3	1 locked, 1:20, layer ASG-VIEWPORT
A3-PORTRAIT	ISO_full_bleed_A3_(297.00_x_420.00_MM)	ASG-TB-A3	1 locked, 1:20, layer ASG-VIEWPORT
A1-LANDSCAPE	ISO_full_bleed_A1_(841.00_x_594.00_MM)	ASG-TB-A1	1 locked, 1:50, layer ASG-VIEWPORT

Each layout: plot area Layout, scale 1:1, centred, plot style table ASG_Monochrome.ctb, plot object lineweights ON, plot with plot styles ON, plot transparency OFF, paper-space last, hide viewport objects OFF. Border: 20 mm left (filing), 10 mm other sides, layer ASG-BORDER. Title blocks are separate designs per size (A4 180 mm, A3 180 mm, A1 250 mm wide) - not stretched. Change a viewport scale: unlock (Properties > Display locked = No), pick scale from status bar, lock again.

Title block attributes: PROJECT, LOCATION, CLIENT, CONSULTANT, DWG_TITLE, DWG_TITLE_2, DWG_NO, REV, DATE, DRAWN_BY, CHECKED_BY, APPROVED_BY, SCALE, SHEET_NO, DWG_STATUS, REMARKS. Edit with EATTEDIT or ATTIPEDIT. Company heading is text - replace with an approved logo block when available (no logo invented).

Plot styles: ASG_Monochrome.ctb: ACI 1-249 & 255 plot black; 250-254 plot as grey 20/36/52/68/84 % (hatch, insulation, reference). Lineweight and linetype = object. Fallback if the CTB is unavailable: AutoCAD's own monochrome.ctb.

PDF: DWG To PDF.pc3, vector quality 1200 dpi, raster 300 dpi, fonts embedded (PDFSHX = 1 / TrueType as text), include layer information OFF for client issue. PDF import (PDFIMPORT) brings geometry on PDF_layers - move to ASG layers and purge.

14. Creating the DWG and DWT (manual - not possible in generation environment)

1) Open ASG_Master_Template.dxf in AutoCAD 2018+. 2) AUDIT (Y). 3) Run APPLOAD > ASG_Template_Setup.lsp > type ASG-TABLESTYLES. 4) SAVEAS > AutoCAD 2018 Drawing (*.dwg) > ASG_Master_Template.dwg. 5) SAVEAS > AutoCAD Drawing Template (*.dwt) > ASG_Master_Template.dwt, description 'ASG master template', Measurement Metric. 6) Copy ASG_Monochrome.ctb to the Plot Styles folder. 7) Re-open the DWT and tick the items in the validation report marked PASS* / UNVERIFIED.

15. Workflow & housekeeping

Topic	Standard
Start	NEW > ASG_Master_Template.dwt only. Never start from acad.dwt.
UCS	Draw in World UCS; UCSVP 1 per viewport; set UCS back to World before saving.
Autosave / recovery	SAVETIME 10, .bak on, local autosave folder; DRAWINGRECOVERY after crash.
Audit / purge	Before issue: AUDIT Y > PURGE all (incl. regapps, orphaned data) > -SCALELISTEDIT reset if bloated > save.
Xrefs	Relative paths only (REFPATHTYPE 1), backgrounds attached as Overlay on ASG-REFERENCE, in project 02-Drawings\Ref; bind only for final archive.
PDF	Underlays on ASG-REFERENCE, PDFIMPORT geometry re-layered to ASG layers.
Compatibility	Save as AutoCAD 2018 DWG (default for AutoCAD 2018-2026); DXF copy for third parties; never save down below 2013 without approval.
File names	Drawing number + short project name, e.g. AS-2026-014-ALU-SD-001-R0_MarinaTower.dwg

Appendix - layout previews (rendered from the saved DXF, monochrome)

A4-LANDSCAPE

SAEED AL SIRAJ GLASS & ALUMINIUM WORKS L.L.C.		
AL SIRAJ GROUP ALUMINIUM & GLASS DIVISION UAE		
PROJECT PROJECT NAME		LOCATION LOCATION, UAE
CLIENT CLIENT NAME		CONSULTANT CONSULTANT NAME
REMARKS -		
DRAWING TITLE DRAWING TITLE		
DRAWN BY -	CHECKED BY -	APPROVED BY -
SCALE AS SHOWN	DATE DD.MM.YYYY	STATUS FOR APPROVAL
DRAWING No. AS-YYYY-NNN-DIS-TYP-001	REV R0	SHEET No. 01 OF 01

A4-PORTRAIT

SAEED AL SIRAJ GLASS & ALUMINIUM WORKS L.L.C.			
AL SIRAJ GROUP ALUMINIUM & GLASS DIVISION UAE			
PROJECT PROJECT NAME		LOCATION LOCATION, UAE	
CLIENT CLIENT NAME		CONSULTANT CONSULTANT NAME	
REMARKS -			
DRAWING TITLE DRAWING TITLE			
-			
DRAWN BY -		CHECKED BY -	
SCALE AS SHOWN		DATE DD.MM.YYYY	
		APPROVED BY -	
		STATUS FOR APPROVAL	
DRAWING No. AS-YYYY-NNN-DIS-TYP-001		REV R0	SHEET No. 01 OF 01

A3-LANDSCAPE

SAEED AL SIRAJ GLASS & ALUMINIUM WORKS L.L.C.			
PROJECT		LOCATION	
PROJECT NAME		LOCATION, UAE	
CLIENT		CONSULTANT	
CLIENT NAME		CONSULTANT NAME	
REMARKS			
.			
DRAWING TITLE			
DRAWING TITLE			
.			
DRAWN BY		CHECKED BY	APPROVED BY
.		.	.
SCALE		DATE	STATUS
AS SHOWN		DD.MM.YYYY	FOR APPROVAL
DRAWING NO.		REV	SHEET NO.
AS-YYYY-NNN-DIS-TYP-001		R0	01 OF 01

A3-PORTRAIT

SAEED AL SIRAJ GLASS & ALUMINIUM WORKS L.L.C.		
AL SIRAJ GROUP ALUMINIUM & GLASS DIVISION UAE		
PROJECT	LOCATION	
PROJECT NAME	LOCATION, UAE	
CLIENT	CONSULTANT	
CLIENT NAME	CONSULTANT NAME	
REMARKS		
DRAWING TITLE		
DRAWING TITLE		
DRAWN BY	CHECKED BY	APPROVED BY
AS SHOWN	DATE	REVISIONS
AS SHOWN	DATE	FOR APPROVAL
DRAWING NO.	REV	DRAWING NO.
AS-YYYY-MMM-DIS-TYP-001	R0	01 OF 01

A1-LANDSCAPE

SAEED AL SIRAJ GLASS & ALUMINIUM WORKS L.L.C.		
SAIRA GROUP / ALUMINIUM & GLASS DIVISION / UAE		
PROJECT NAME	LOCATION, UAE	
CLIENT NAME	CONSULTANT NAME	
DRAWING TITLE		
1		
DESIGNED BY	REVIEWED BY	APPROVED BY
AS SHOWN	DELIVERY	FOR APPROVAL
AS-YYY-NNN-DIS-TYP-001	RD	01 OF 01